Al Isra Private University
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Electrical Engineering Department

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Second Exam

Course Name:	Systems & Signals Processing	Date:	Jan. 3, 2005
Course No:	402-322	ExamTime:	1
Instructor:	Eng. Juma Abu-Ghalune	No. of pages:	1

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 4.exp(-4t).sin(400\pi t)u(t)$.
   b. $y(t) = (1/t).cos(200\pi t)$.
   c. $z(t) = 8/(4 + j200\pi t)^2$.

2. Using Table 4-1 and Table 4-2, find the Fourier Transform Inverse of:
   a. $X(f) = 3.sinc(4f - 8)$.
   b. $Y(f) = 5.exp(j20\pi f).\Lambda(0.2f)$.
   c. $Z(f) = 8/|4 + j8\pi f|^2$.

3. Use the differentiation theorem to find the Fourier transform of the the signal $x(t) = r(t + 2) - 2.r(t + 1) + 2.r(t - 1) - r(t - 2)$.

4. Consider a fixed linear system with amplitude response $2.\Pi[w/600\pi] + 4.\Pi[w/400\pi]$ and phase response $-j0.002\ w$ for $|w| < 250\pi$, 0.5π for $w < -250\pi$, and -0.5π for $w > 250\pi$. Find the output and determine the type of distortion for the following inputs:
   a. $w(t) = cos(250\pi t) + 4\ cos(275\pi t)$.
   b. $x(t) = cos(50\pi t) + 2\ cos(150\pi t)$.
   c. $y(t) = cos(150\pi t) + cos(250\pi t)$.

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Q1 a. $x(t) = 4e^{-4t} \sin(400\pi t)\, u(t)$ [6]

$$X(f) = \frac{4}{4 + j2\pi f} * \frac{1}{2j}(\delta(f-200) - \delta(f+200))$$

$$= \frac{+j2}{4 + j2\pi(f+200)} - \frac{j2}{4 + j2\pi(f-200)}$$

b. $y(t) = \frac{1}{t} \cdot \cos(200\pi t)$

$$Y(f) = -j\pi \operatorname{sgn}(f) * \frac{1}{2}[\delta(f-100) + \delta(f+100)]$$

$$= -j\frac{\pi}{2}[\operatorname{sgn}(f-100) + \operatorname{sgn}(f+100)]$$

c. $z(t) = \frac{8}{(4 + j200\pi t)^2} = \frac{8}{(100)^2[0.04 + j2\pi t]^2}$

$$Z(f) = \frac{8}{(100)^2} \cdot (-f)\, e^{0.04f}\, u(-f).$$

Q2 a. $X(f) = 3\,\text{sinc}(4(f-2)) = \frac{3}{4}(4\,\text{sinc}(4(f-2)))$ [6]

$$x(t) = \frac{3}{4} e^{j4\pi t}\, \Pi\left(\frac{t}{4}\right).$$

b. $Y(f) = 5.e^{j2\pi(10)f}\, \Lambda(0.2f) = 5\, e^{j2\pi(10)f}\, \Lambda\left(\frac{f}{5}\right)$

$$y(t) = 5.5\,\text{sinc}^2(5(t+10)) = 25\,\text{sinc}^2(5(t+10)).$$

c. $Z(f) = \frac{8}{|4 + j8\pi f|^2} = \frac{8}{(4)^2 + (4)^2(2\pi f)^2}$

$$= \frac{8}{16} \frac{1}{(1 + (2\pi f)^2)} = \frac{1}{4} \frac{2}{(1 + (2\pi f)^2)}$$

$$z(t) = 0.25\, e^{-|t|}$$

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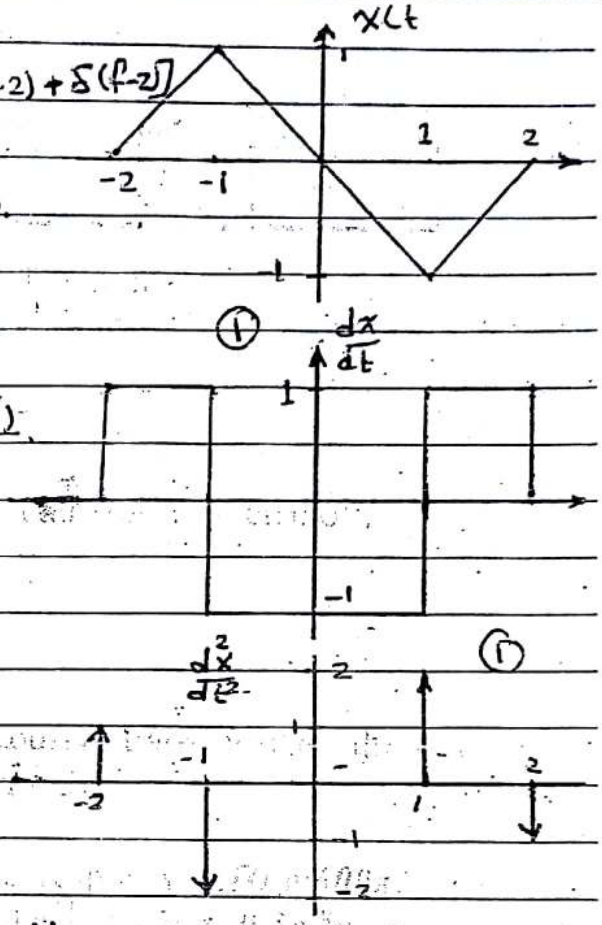
(5) $\frac{d^2x}{dt^2} = 2[\delta(t-1) - \delta(t+1)] + 1[\delta(f+2) + \delta(f-2)]$

$F\{\frac{d^2x}{dt^2}\} = j4\sin(2\pi f) + j2\sin(4\pi f)$.

$= (j2\pi f)^2 \; X(f)$.

$X(f) = -j\frac{\sin(2\pi f)}{(\pi f)^2} + j\frac{1}{2}\cdot\frac{\sin(4\pi f)}{(\pi f)^2}$

$X(f) = X(-f)$



Q.4 a. $x(t) = \cos(250\pi t) + 4\cos(275\pi t)$

10 $w_o(t) = 2\cos(250\pi t - \frac{\pi}{2}) + 8\cos(275\pi t - \pi/2)$.

No distortion.

b. $x(t) = \cos(50\pi t) + 2\cos(150\pi t)$

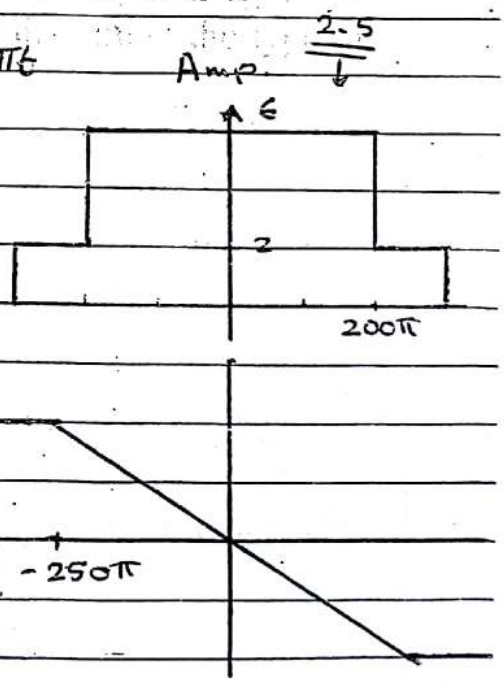
$x_o(t) = 6\cos(50\pi t - 0.1\pi) + 12\cos(150\pi t - 0.3\pi)$.

No distortion.

c. $y(t) = \cos(150\pi t) + \cos(250\pi t)$

$y_o(t) = 6\cos(150\pi t - 0.3\pi) + 2\cos(250\pi t - \frac{\pi}{2})$

Amplitude distortion.



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